

about

padajar-templates is a series of $\text{\LaTeX}$ classes made by Phi Adajar (that's me!). This originated as a series of $\text{\LaTeX}$ commands in summer 2021, used just for psets, but has since transformed into several classes (padajar-memo, padajar-paper, padajar-pset, and padajar-slides) relying on a common package (padajar-defaults). I'm hoping the up-front cost of creating them will be paid back many times over in my future work, and that of others.

These files are open-source and distributed under an MIT License (see [Appendix C](#)). If you do use them, I greatly appreciate hearing about it, attribution, contributing (if that's up your alley), and spreading the word!

contents

1	padajar-defaults	2
2	classes	3
2.1	padajar-memo	3
2.2	padajar-paper	4
2.3	padajar-pset	5
2.4	padajar-slides	6
3	issues	6
4	planned features	8
5	unplanned features	8
A	changelog	9
B	installation instructions	9
C	license	10

1 padajar-defaults

All `padajar-templates` load the package `padajar-defaults`. This package comes with the following features/options.

- Body and math font set to IBM Plex. Body text is IBM Plex Sans by default, but can be changed with the `[serif]` option to IBM Plex Serif. Math font is always serif, like $\sin^2(\alpha) + \cos^2(\alpha) = 1$ and

$$\Phi(x) = \int_{-\infty}^x \frac{e^{-t^2/2}}{\sqrt{2\pi}} dt.$$

Mono font is set to IBM Plex Mono (like `so`). All fonts are set to user a lighter weight than the default.

Note that because the `fontspec` package is used, to use this package (and any below templates), **you must** use XeLaTeX or LuaLaTeX. (For me, compilation with the former has been slightly faster.) You can always set this by adding `% !TEX program = xelatex` (without quotes) to the top of your document (i.e., before `\documentclass`).

- A host of packages loaded by default:
 - for math: `amsmath`, `amsthm`, `physics`, `bbm`...
 - for graphics: `graphicx`, `pdfpages`, `listings`...
 - for tables: `tabularx`, `booktabs`, `threeparttable`...
 - for general utility: `hyperref`, `cleveref`, `enumitem`, `xcolor`...

If you want the full list, head into the source code of this package.

- A `listings` definition for Stata (modified from [here](#)), along with a nicer `listings` scheme.
- Some math commands I find useful: `\eps` (ε), `\E` (E), `\V` (V), `\indic` ($\mathbb{I}$), `\Pr` (P), `\Cov` (Cov), `\Var` (Var), and `\indep` ($X \perp\!\!\!\perp Y$).
- Table helpers: `\sym` for `esttab` significance stars (works in and out of math mode), L/R/C `tabularx` columns (ragged-right, ragged-left, and centered X columns), and an expandable `\input` inside `tabular`, so `esttab` fragments can be `\input` into a table.
- Figure captions sit above the graphic, and the space between floats and body text is tightened to 10pt.
- A `padajar` color scheme! :)

A few other options are used internally by the classes to differentiate between documents and slides.

Notably, `padajar-defaults` is a standalone package. It can be used in any document, not just those in `padajar-templates`, granting additional flexibility.

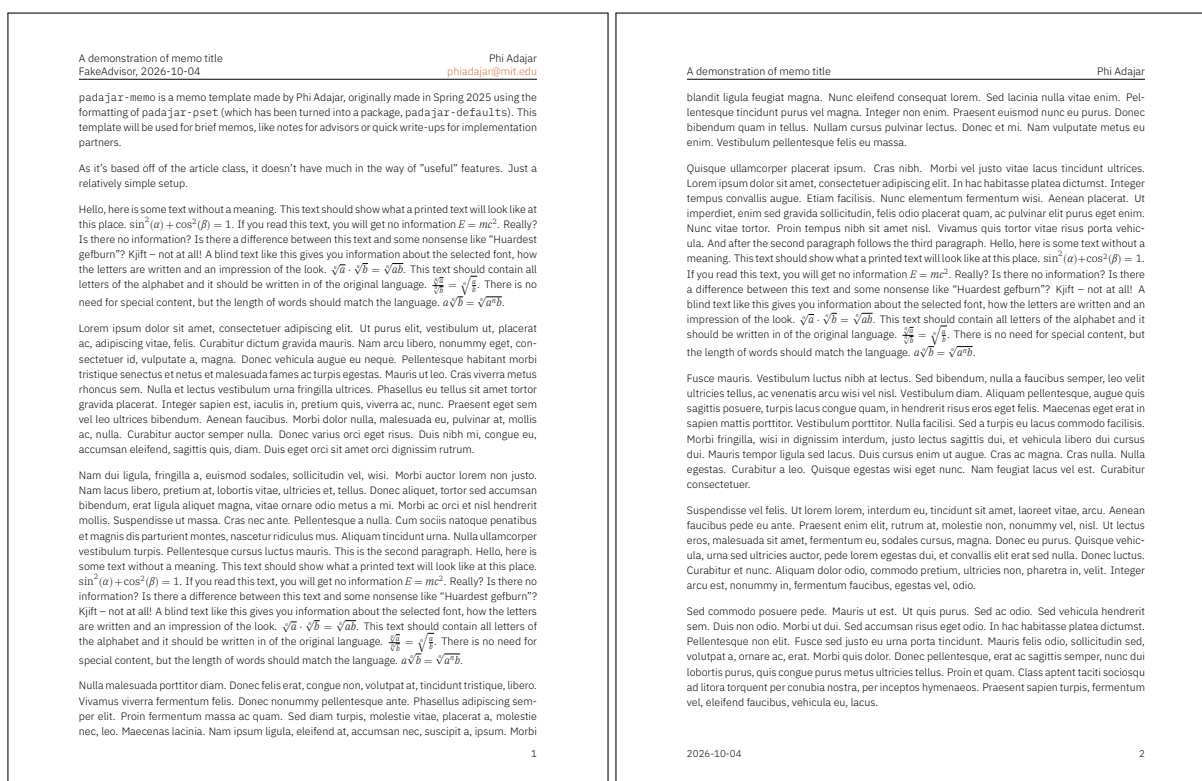
2 classes

I've made four different formats that use padajar-defaults, each with its own purpose. These cover the vast majority of things that I create: paragraph-dense writing, papers, problem sets, and presentation slides. I've put the same design ethos through them (and my website), with the intent of making my digital presence a tad more “cohesive”.

2.1 padajar-memo

This is the simplest template of the four, and is the one that you're currently viewing.

Figure 1: A preview of the padajar-memo class from memo-example.pdf.



padajar-memo is intended for ideas that are communicated with paragraphs, tables, and figures. At its core, it's really just an article document with my usual settings. It comes with these additional features:

- Fancy headers and footers, which take information from `\name`, `\email`, `\date`, `\memotitle`, and `\memonote`.
- Commands for adding in-line self-notes and alerts, which I use for to-dos and other comments. these commands are `\selfnote` (purple and bold) and `\alert` (orange and bold).
- Analogously, the environments `\Selfnote` and `\Alert` can contain multi-paragraph entries with the same formatting.

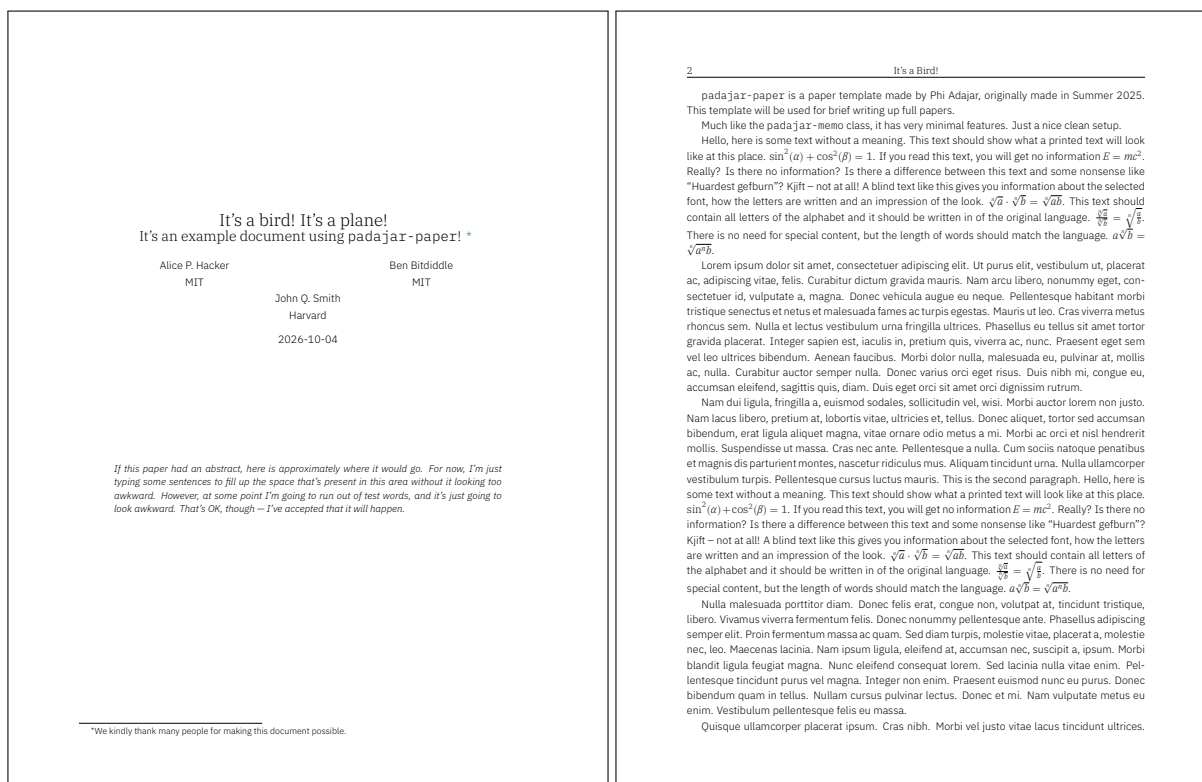
- Theorem-like environments (theorem, lemma, proposition, corollary, definition, example, remark), plus assumption and implication on their own counters. All of these, and proof, are indented 0.5in on both sides, and the paragraph after them is not indented.
- Automatic appendix formatting: after `\appendix`, tables, figures, and theorems renumber as A.1, and `\cref` and `\Cref` print sections as “Appendix A”.
- An empty `\memonote{}` drops the note from the first-page header, leaving only the date.

...and that’s really it. The base setup of this class isn’t too far different from the `article` class. It’s just meant to be a clean baseline template so I don’t always have to futz around with the same imports and options that I always do.

2.2 padajar-paper

This minor variant of the `padajar-memo` document class is to be used for writing up papers. There’s not much different about this besides some nice formatting.

Figure 2: A preview of the `padajar-paper` class from `paper-example.pdf`.



Some notable features include:

- Formatted title pages using `\maketitle` (the former `\makepapertitle` is retained as an alias). This will load `\papertitle`, `\papershorttitle`, and optionally, `\paperthanks`

and `\paperabstract`. Authors are formatted using the `authblk` package, as name-over-affiliation blocks in equal-width centered columns. `\paperpdfauthors` sets the author list in the PDF metadata.

- `\usdate` prints a US-style date (“October 4, 2026”), and `\latestversion{url}` adds a “Click here for the latest version” link on its own line under the date. Use them together as

```
\date{\usdate \latestversion{url}}
```

- Indented paragraphs with no space between them, and AER-style references (smaller font, tight entries). The repository includes `aea.bst` for `\bibliographystyle{aea}`.
- A “two-sided” document format, with headers that alternate between `\papershortauthors` and `\papershorttitle`.

Otherwise, it’s identical to the `padajar-memo` class.

2.3 padajar-pset

This class is meant for writing up psets or exams with solutions. As a modification of the exam class, it retains all exam class functionality, like `parts` and `subparts`, the `solution` environment, and creating versions with and without answers included. For more information, refer to [this Overleaf page](#).

Some key features I’ve added in:

- The `reflection` environment. These question reflections are intended to help connect questions to the course as a whole, interesting insights gained through the question, and more. Really, they’re just a way to help think more about the “why” of questions. These are encased in a gray `colorbox`, and can be included either by passing the `reflections` option to the class or adding `printreflections` to the preamble.
- The `solution` environment is encased in a green `tcolorbox`.
- Headers and footers which take information from `\name`, `\email`, `\date`, `\classnum`, `\classname`, and `\assignment`. Optionally, `\professors` and `\collaborators` can take infinite arguments (separated with curly brackets) and add that information to the first page.

And some usage notes:

- Questions are styled like sections; you should use `\titledquestion` command for creating questions, and *then* write your question text (i.e., not in the question title). While I know this won’t work for every field, multi-part questions are the bread and butter of economics psets.
- You can’t use `table` environment (or any other `float` environment) inside the `solution` environment. This is currently solved using the `float` package, and then using the command `\begin{table}[H]`. I’m still looking for a “better” solution to this.

Figure 3: A preview of the padajar - pset class from pset-example.pdf.

14.XX: Assignment Name Phi Adajar

Solving the first and second equations for λ , we get

$$\lambda = \frac{1}{3} \frac{1}{p_A} \left(\frac{B}{A} \right)^{\frac{1}{2}} = \frac{2}{3} \frac{1}{p_B} \left(\frac{A}{B} \right)^{\frac{1}{2}} \Leftrightarrow p_B B = 2 p_A A$$

indicating that the consumer will spend twice their income on B than they will on A .
If we desire $A = 2B$, we must then have that $p_B = 4 p_A$.

Reflection. Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetur adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Problem 2. Example: 14.381 Fall 2021 PS1, #2. (10 points)

This problem set question is from 14.381, Statistical Methods in Economics, taken with Whitney Newey. Consider the gasoline demand data that is provided on Canvas and two OLS regressions (all in levels, not logs) for that data:

- the OLS regression of the gasoline purchases on a constant, price, and income
- the OLS regression of the level of gasoline purchases on a constant, price, income, and covariates consisting of the average age of drivers in the household, the number of drivers in the household, and the dummy variable for the availability of public transport.

Assume that there is no heteroskedasticity or autocorrelation.

(3 points) (a) Give Tables of OLS estimates and standard errors for all the coefficients for both regressions i) and ii), assuming there is no heteroskedasticity or autocorrelation.

Solution. We use the following code:

```
1 import delimitated "Gasoline_data.csv"
2
3 zencode (v1 v2 v3 v4 v5 v6 v7 v8 v9 v10)
4 (state log_quantity log_price log_income dist_gulf_of_mexico ///
5  log_drivers public_transit mean_age_drivers log_price_cents ///
6  state_gas_tax)
7
8 foreach v of varlist quantity price income drivers {
9   gen 'v' = exp(log_'v')
10 }
11
12 eststo clear
13 eststo: reg quantity price income
14 eststo: reg quantity price income drivers mean_age_drivers public_transit
15
16 esttab
17 esttab using psi-2a.tex, se replace
```

which yields the following table:

Page 4 of 5

14.XX: Assignment Name Phi Adajar

	(1)	(2)
price	quantity -3.359*** (1.006)	quantity -2.500* (0.987)
income	0.000*** (0.000)	0.000*** (0.000)
drivers	2.241*** (0.107)	
mean_age_drivers	-0.023*** (0.005)	
public_transit	-0.643*** (0.185)	
_cons	7.856*** (1.336)	3.986** (1.366)
N	8908	8908

Standard errors in parentheses
* p < 0.05, ** p < 0.01, *** p < 0.001

(3 points) (b) Do an F-test of the null hypothesis that all the covariates have zero coefficients.

Solution. Using

```
1 test drivers mean_age_drivers public_transit
```

after the above command yields $F(3, 8902) = 201.24$, with a p-value of 0.000. This indicates that the covariates are significantly distinct from 0 at the 5% level.

(4 points) (c) Do a t-test of the null hypothesis that the short regression (a) price coefficient is the same as the long regression (b) price coefficient. You may assume that the variance of the difference of the long and short regression coefficients is the difference of their variances. What do you conclude from this test? Does this test lead to a different conclusion than the test in (b)?

Solution. Note that we can approximate this with a z-test because of the size of the sample, which gives a difference of 0.859 and a standard error of $\sqrt{1.006^2 - 0.987^2} = 0.195$. This gives a z-value of 4.414, and a p-value of 0.000, indicating that the two coefficients on price are statistically distinct at the 5% level.

This is the conclusion we would expect from (b); as the coefficients had significant impacts on the regression, we expect them to change the size of the coefficient on price.

Page 5 of 5

2.4 padajar-slides

This class is based on the `moloch` theme for the `beamer` class, hosted on CTAN [here](#).¹ These are used for presentations, including both paper talks and lecture slides.

Key features include:

- Navigation bubbles with section titles on the bottom of slides
- An optional argument for the class, `[sectionslides]`, which adds a navigation slide at the start of each section.
- `\graycitep`, which will cite bibliography items like `\citep` from the `natbib` package, but will insert citations into smaller, gray text.
- `\btLstHL`, which can highlight specific lines in an `lstlisting` environment by using the option `linebackgroundcolor=\btLstHL<overlay spec>\{range list\}`
- A default 16:9 aspect ratio, the objectively correct aspect ratio for presentations.

3 issues

This set of files is *far* from perfect right now. Some high-priority issues on my to-do list include:

¹`moloch` itself is a fork of the `metropolis` theme, hosted on CTAN [here](#). `metropolis` is no longer maintained, and `moloch` was created to fix some of its bugs — see [this blog post](#).

Figure 4: A preview of the padajar-slides class from slides-example.pdf.

a slide template

the padajar-slides class

phi adajar (they/them)
phiadajar@mit.edu

v1.5.0

2026-10-04

guiding philosophy:
you want people to **understand**.
design with that in mind.

Engagement is strongest predictor of take-up

What closes the achievement gaps for top students?

Table 1: Access (%) by selection criteria used

	(1)	(2)	(3)	(4)	(5)
Selected w/ Admin	0.0678 (3.743)				-1.625 (3.920)
Selected w/ Endline		1.233 (3.492)			-1.073 (3.553)
Selected w/ Engagement			8.019* (3.391)		6.645 (3.446)
Selected w/ Grit				-9.489 (7.068)	-7.860 (8.326)
Cons.	61.84*** (3.151)	61.14*** (2.680)	58.07*** (2.347)	62.77*** (1.750)	61.14*** (4.727)
N	1464	1464	1464	1464	1464

Standard errors in parentheses. *p<0.05, **p<0.01, ***p<0.001.

- Access to mentors greatly enhances participation, performance (Ellison and Swanson, 2016; Calaway, 2024)
- Intensive, targeted 6-week program at an elite STEM university shifted students towards elite schools (12pp increase in BA attainment), with expected earnings effects of 3-15% (Cohodes et al., 2024)
- Self-paced classes and tutoring have proven useful for helping students across the ability distribution, and potentially highly scalable (Carlana and Ferrara, 2021; Muralidharan et al., 2019; Koedinger et al., 1997)

- Missing characters. I've done **some shenanigans** to make the math font thinner, and as a result, there are several missing math characters: `\epsilon`, `\varpi`, `\varrho`, `\nabla`, `\surd`, `\mapsto`, `\lmoustache`, `\rmoustache`, `\arrowvert`, and `\Arrowvert`. The first is set to display as `\varepsilon` (ϵ), which is what I primarily use anyways, and the last two are legacy and shouldn't pose much of an issue. These should all be manually set to other fonts, but because these don't see much use for me, I haven't found the best replacements yet. One I have already redefined is `\partial`, which is lifted from T_EX Gyre Schola; peek inside `padajar-defaults` to see how missing characters should be manually defined.
- Find a better solution for that pesky interaction between floats and `\tcolorbox`.
- Fix a few things I *know* are bad L^AT_EX coding:
 - When the body text is sans-serif, the math text also becomes sans-serif. To change it back to serif, I use `\renewcommand{\text}[1]{\oldtext{\rmfamily #1}}`. Yuck.
 - Yeah, yeah, I know I should fix `\epsilon` because *some* people want to use it.
 - Refactor the code, particularly looking for any repeated code and cleaning up the margins/parskip/etc. formatting; that in particular has become a bit of a jumble.
 - Nice code for coloring specific rows/cells of a table? Perhaps something like **here** or **here**.

4 planned features

These are features I'd like to add in the future, but aren't high priority for me at the moment.

- Create the `padajar-notes` class. I'd essentially like to make something with the aesthetics of the other document classes here, but with functionality more akin to [lindrew's](#) notes package.
- Make `\btLstHL` work outside of beamer classes (i.e., in documents).
- More common math shortcuts? I know there's plenty more I use.
- Add the `subfiles` package (pending research; some online seem to strongly prefer just using `\include` and `\includeonly`).
- A much closer look at all math symbols, finding alternatives for ones that I think have too much weight.
- Just a general look into compile time, and seeing if there are any ways to speed it up.

5 unplanned features

Conversely, here are some things that I explicitly *don't* plan on doing with this set of classes.

- Adding these to CTAN. I think it's incredibly unlikely that these see widespread usage, and probably isn't worth my time. I also think there's far too much idiosyncrasy in all of these formats for them to be used widely. But I guess if you think this has a broader interest, let me know?
- Letting fonts, by default, be something other than IBM Plex. I just like the aesthetic. You're more than welcome to do what you want to these files, of course. I just won't be helping you.

A changelog

- 1.5.0 (2026-10-04) figure captions above the graphic with tighter float spacing; `\Cov`, `\Var`, `\indep`; `\sym` and R/L/C columns moved to `padajar-defaults`; equation references print with parentheses; assumption and implication environments; theorem-like environments and proofs indented 0.5in; appendix numbering moved into `padajar-memo`; empty `\memonote` shows only the date; `padajar-paper`: author blocks in equal-width columns, `\usdate`, `\latestversion`, `\paperpdfauthors`, indented paragraphs, AER-style references, `aea.bst` included; footnote text keeps the article layout
- 1.4.0 (2026-07-11) add `\Pr` (blackboard $\mathbb{P}$); automatic appendix formatting in `padajar-paper` (A.1 table/figure/theorem numbering and “Appendix A” cross-references); upright equation references; expandable `\input` inside `tabular` for `esttab` fragments
- 1.3.1 (2026-07-11) page-geometry cleanup: `\headheight` sized to the header and `\textheight` recomputed through `geometry`, with consistent page margins across `padajar-memo`, `padajar-paper`, and `padajar-pset`; flexible `\parskip` for more even page fill; footnotes pinned to the page bottom via `footmisc`
- 1.3.0 (2026-07-11) paper title pages now use `\maketitle` (`\makepapertitle` kept as a backwards-compatible alias); restore `\sym`; sharper `\indic` (via `dsfont`); add `cleveref` names and R/L/C `tabularx` columns; fix the `serif` option across classes; misc header/typesetting fixes
- 1.2.0 (2025-09-07) add `\padajar-paper`, other minor typesetting
- 1.1.0 (2025-06-25) add `\alert` and `\selfnote`, minor fixes
- 1.0.0 (2025-03-30) first creation of the `padajar-memo`, `padajar-pset`, and `padajar-slides` classes, each importing the `padajar-defaults` package
- 0.1.0 (2021-07-19) initial creation of what would become these classes: a series of $\text{\LaTeX}$ commands accessed with deprecated `\input` command

B installation instructions

To use this package and set of classes:

1. Download the following files:

- `padajar-defaults.sty`
- `padajar-memo.cls`
- `padajar-paper.cls`
- `padajar-pset.cls`
- `padajar-slides.cls`

- `aea.bst` (optional; AER bibliography style for papers)
2. Place them into *one* of the following places:
 - The same folder as the `.tex` that you are trying to create. This will make it available for only that project.
 - Add it to a local $\text{\TeX}$ folder, where instructions will vary depending on your $\text{\TeX}$ installation. This will make it available for all projects on your system.
 3. Compile your document using XeLaTeX or LuaLaTeX (required due to `fontspec`).

C license

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